



4425 Manchester Rd

Kalamazoo, MI 49001

Phone 269 381-9666

Fax 269 381-9698

www.karlabs.com

**City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601**

KAR Project No. : 703913

Date Reported : 04/12/17

Date Activated : 03/29/17

Date Due : 04/12/17

Date Validated : 04/12/17

Attn : Ms. Cindy Tomaszewski

Project

Description : Analysis of one Spring Biosolids sample.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including the Maximum Contaminant Level (MCL) set by the USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 703913. To arrange additional sampling or testing please contact our Client Services Department.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, reading "David R. Alkema". The signature is written in a cursive style with a large, stylized "D" and "A".

David R. Alkema
Laboratory Manager

KAR Laboratories, Inc. maintains Full Certification status for Bacteriology, Inorganics, Regulated Organics and Synthetic Organics through USEPA, Michigan Department of Public Health and Indiana State Department of Health. This report may only be reproduced in full and not without the written consent of **City of Cadillac, Water Resources**.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **703913**

Date Reported : **04/12/17**

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : <u>"Digester #4"</u> Sampled By : <i>DS of City of Cadillac, Water Resources</i> Sample Date : <i>03/28/17</i> Sample Time : <i>1330</i>						
				Date Received : <i>03/29/17</i> Sample Type : <i>sludge</i> KAR Sample No. : 703913-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep. Hg	Completed		EPA 7471A	04/07/17	MID	
Prep. metals	Completed		EPA 200.7/8	03/30/17	MID	
Aluminum, total	4200	mg/kg total solids	EPA 200.8	04/03/17	NHM	Detection limit = 99
Arsenic, total	26.6	mg/kg total solids	EPA 200.8	04/03/17	NHM	Detection limit = 5.0
Cadmium, total	<2.5	mg/kg total solids	EPA 200.8	04/03/17	NHM	
Calcium, total	29,300	mg/kg total solids	EPA 6010B	04/07/17	JHB	Matrix effect observed; result is approximate. Detection limit = 50
Chromium, total	97.2	mg/kg total solids	EPA 200.8	04/03/17	NHM	Detection limit = 5.0
Copper, total	1020	mg/kg total solids	EPA 200.8	04/03/17	NHM	Matrix effect observed; result is approximate. Detection limit = 20
Lead, total, by ICP	31.4	mg/kg total solids	EPA 200.8	04/03/17	NHM	Detection limit = 25
Magnesium, total	5220	mg/kg total solids	EPA 6010B	04/07/17	JHB	Detection limit = 50
Mercury, total	0.35	mg/kg total solids	EPA 7471A	04/07/17	MID	Detection limit = 0.12
Molybdenum, total	25.0	mg/kg total solids	EPA 200.8	04/03/17	NHM	Detection limit = 9.9
Nickel, total	29.3	mg/kg total solids	EPA 200.8	04/03/17	NHM	Detection limit = 9.9
Potassium, total	4170	mg/kg total solids	EPA 6010B	04/07/17	JHB	Detection limit = 50
Selenium, total	6.3	mg/kg total solids	EPA 200.8	04/03/17	NHM	Detection limit = 2.5
Silver, total	4.0	mg/kg total solids	EPA 200.8	03/31/17	NHM	Detection limit = 2.0
Sodium, total	6200	mg/kg total solids	EPA 6010B	04/07/17	JHB	Detection limit = 500
Zinc, total	1480	mg/kg total solids	EPA 200.8	04/03/17	NHM	Detection limit = 20
TCLP extraction	Completed		EPA 1311	03/30/17	JWW	
TCLP report	See comment			04/12/17	KMC	This material does not exhibit the Toxicity Characteristic with respect to the requested TC parameters.
ZHE extraction	Completed		EPA 1311	03/30/17	JWW	
Chloride	8680	mg/kg total solids	EPA 300.0A	04/12/17	ALK	Detection limit = 200
Nitrogen, ammonia	38,100	mg/kg total solids	EPA 350.1	04/04/17	JWW	Detection limit = 800
Nitrogen, nitrate	<20	mg/kg total solids	EPA 300.0A	04/12/17	ALK	
Nitrogen, total kjeldahl	91,200	mg/kg total solids	EPA 351.2	04/05/17	JWW	Detection limit = 5800
Phosphorus, total (as P)	40,400	mg/kg total solids	SM 4500-P B5,E	04/05/17	EIF	Detection limit = 8000
Sulfate	8350	mg/kg total solids	EPA 300.0A	04/12/17	ALK	Detection limit = 400
Waste pH	8.1	S.U.	EPA 9045C	04/10/17	JWW	
Solids, total	2.52	% by weight	EPA 160.3	03/30/17	LPK	Detection limit = 0.01
Solids, volatile	57.20	% of total solids	EPA 160.4	03/30/17	LPK	Detection limit = 0.01
Prior. Poll. volatiles	See below		EPA 8260	04/03/17	JAR	
Prep. VOA	Completed		EPA 5030	04/03/17	JAR	
1,1,1-Trichloroethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
1,1,2,2-Tetrachloroethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
1,1,2-Trichloroethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
1,1-Dichloroethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
1,1-Dichloroethene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
1,2-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	

Attest: 
David R. Alkema, Lab Manager

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **703913**

Date Reported : **04/12/17**

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID <u>"Digester #4"</u> : Sampled By : <i>DS of City of Cadillac, Water Resources</i> Sample Date : <i>03/28/17</i> Sample Time : <i>1330</i>						
				Date Received : <i>03/29/17</i> Sample Type : <i>sludge</i> KAR Sample No. : 703913-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,2-Dichloroethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
1,2-Dichloropropane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
1,3-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
1,4-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
2-Chloroethylvinyl ether	<20,000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Acrolein	<40,000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Acrylonitrile	<4000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Benzene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Bromodichloromethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Bromoform	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Bromomethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Carbon tetrachloride	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Chlorobenzene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Chloroethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Chloroform	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Chloromethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Cis-1,3-Dichloropropene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Dibromochloromethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Ethylbenzene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Methylene chloride	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Tetrachloroethene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Toluene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Trans-1,2-Dichloroethene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Trans-1,3-Dichloropropene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Trichloroethene	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Trichlorofluoromethane	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Vinyl chloride	<2000	ug/kg total solids	EPA 8260	04/03/17	JAR	
Prior. Poll. acids	See below		EPA 8270	03/30/17	GMB	
Prior. Poll. base-neutrals	See below		EPA 8270	03/30/17	GMB	
Prep. SV Acid/BN	Completed		EPA 3545	03/30/17	GMB	
1,2,4-Trichlorobenzene 8270	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
1,2-Dichlorobenzene by 8270	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
1,2-Diphenylhydrazine	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
1,3-Dichlorobenzene by 8270	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
1,4-Dichlorobenzene by 8270	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2,3,7,8-TCDD by 8270	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2,4,6-Trichlorophenol	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2,4-Dichlorophenol	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2,4-Dimethylphenol	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2,4-Dinitrophenol	<26,000	ug/kg total solids	EPA 8270	03/30/17	GMB	

Attest: 
David R. Alkema, Lab Manager

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **703913**

Date Reported : **04/12/17**

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : <u>"Digester #4"</u>						
Sampled By : DS of City of Cadillac, Water Resources				Date Received : 03/29/17		
Sample Date : 03/28/17				Sample Type : sludge		
Sample Time : 1330				KAR Sample No. : 703913-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
2,4-Dinitrotoluene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2,6-Dinitrotoluene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2-Chloronaphthalene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2-Chlorophenol	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2-Methyl-4,6-dinitrophenol	<26,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
2-Nitrophenol	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
3,3'-Dichlorobenzidine	<26,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
4-Bromophenyl phenyl ether	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
4-Chloro-3-methylphenol	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
4-Chlorophenyl phenyl ether	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
4-Nitrophenol	<26,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Acenaphthene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Acenaphthylene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Anthracene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Benzidine	<65,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Benzo(a)anthracene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Benzo(a)pyrene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Benzo(b)fluoranthene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Benzo(ghi)perylene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Benzo(k)fluoranthene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Bis(2-chloroethoxy)methane	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Bis(2-chloroethyl)ether	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Bis(2-chloroisopropyl)ether	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Bis(2-ethylhexyl)phthalate	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Butylbenzyl phthalate	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Chrysene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Di-n-butyl phthalate	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Di-n-octyl phthalate	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Dibenzo(ah)anthracene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Diethyl phthalate	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Dimethyl phthalate	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Fluoranthene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Fluorene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Hexachlorobenzene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Hexachlorobutadiene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Hexachlorocyclopentadiene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Hexachloroethane	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Indeno(123cd)pyrene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Isophorone	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
N-Nitrosodi-n-propylamine	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	

Attest: 
David R. Alkema, Lab Manager

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **703913**

Date Reported : **04/12/17**

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID	<u>"Digester #4"</u>					
:						
Sampled By :	DS of City of Cadillac, Water Resources			Date Received :	03/29/17	
Sample Date :	03/28/17			Sample Type :	sludge	
Sample Time :	1330			KAR Sample No. :	703913-01	
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
N-Nitrosodimethylamine	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
N-Nitrosodiphenylamine	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Naphthalene by Method 8270	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Nitrobenzene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Pentachlorophenol	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Phenanthrene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Phenol	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Pyrene	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Prior. Poll. pest./PCB's	See below		EPA 8270	03/30/17	GMB	
4,4'-DDD	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
4,4'-DDE	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
4,4'-DDT	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Aldrin	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Alpha-BHC	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Beta-BHC	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Chlordane	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Delta-BHC	<13,000	mg/kg total solids	EPA 8270	03/30/17	GMB	
Dieldrin	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Endosulfan I	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Endosulfan II	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Endosulfan sulfate	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Endrin	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Endrin aldehyde	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Gamma-BHC	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Heptachlor	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Heptachlor epoxide	<13,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
PCB Aroclor 1016 by 8270	<130,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
PCB Aroclor 1221 by 8270	<130,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
PCB Aroclor 1232 by 8270	<130,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
PCB Aroclor 1242 by 8270	<130,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
PCB Aroclor 1248 by 8270	<130,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
PCB Aroclor 1254 by 8270	<130,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
PCB Aroclor 1260 by 8270	<130,000	ug/kg total solids	EPA 8270	03/30/17	GMB	
Toxaphene	<130,000	ug/kg total solids	EPA 8270	03/30/17	GMB	

Attest: 
David R. Alkema, Lab Manager

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **703913**

Date Reported : **04/12/17**

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : **TCLP Leachate of Digester #4**

Sampled By :

Date Received : **03/29/17**

Sample Date :

Sample Type : **TCLP**

Sample Time :

KAR Sample No. : **703913-01T**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Metals	Completed		EPA 200.7	03/31/17	JHB	
Prep. Hg	Completed		EPA 7470A	04/05/17	MID	
Prep. metals	Completed		EPA 200.7/8	03/31/17	MID	
Arsenic, total, by ICP	<0.5	mg/L	EPA 200.7	03/31/17	JHB	TC regulatory limit is 5.0 mg/L.
Barium, total	0.63	mg/L	EPA 200.7	03/31/17	JHB	TC regulatory limit is 100 mg/L.
Cadmium, total	<0.025	mg/L	EPA 200.7	03/31/17	JHB	TC regulatory limit is 1.0 mg/L.
Chromium, total	<0.05	mg/L	EPA 200.7	03/31/17	JHB	TC regulatory limit is 5.0 mg/L.
Copper, total	<0.1	mg/L	EPA 200.7	03/31/17	JHB	
Lead, total, by ICP	<0.25	mg/L	EPA 200.7	03/31/17	JHB	TC regulatory limit is 5.0 mg/L.
Mercury, total	<0.05	mg/L	EPA 7470A	04/05/17	MID	TC regulatory limit is 0.2 mg/L.
Selenium, total, by ICP	<0.5	mg/L	EPA 200.7	03/31/17	JHB	TC regulatory limit is 1.0 mg/L.
Silver, total	<0.025	mg/L	EPA 200.8	03/31/17	NHM	TC regulatory limit is 5.0 mg/L.
Zinc, total	0.98	mg/L	EPA 200.7	03/31/17	JHB	
TC Semi-Volatiles	See below		EPA 8270	03/31/17	GMB	
Prep. SV Acid/BN	Completed		EPA 3510	03/31/17	MHK	
Prep. herbicides	Completed		EPA 8151	03/31/17	GMB	
2,4,5-Trichlorophenol	<0.1	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 400 mg/L.
2,4,6-Trichlorophenol	<0.1	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 2.0 mg/L.
2,4-Dinitrotoluene	<0.1	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 0.13 mg/L.
Cresols	<0.2	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 200 mg/L.
Hexachlorobenzene	<0.1	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 0.13 mg/L.
Hexachlorobutadiene	<0.1	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 0.5 mg/L.
Hexachloroethane	<0.1	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 3.0 mg/L.
Nitrobenzene	<0.1	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 2.0 mg/L.
Pentachlorophenol	<0.1	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 100 mg/L.
Pyridine	<0.2	mg/L	EPA 8270	03/31/17	GMB	TC regulatory limit is 5.0 mg/L.
TC Pesticides by ECD	See below		EPA 8081	03/31/17	GMB	
Prep. ECD	Completed		EPA 3510	03/31/17	GMB	
Chlordane	<0.02	mg/L	EPA 8081	03/31/17	GMB	TC regulatory limit is 0.03 mg/L.
Endrin	<0.004	mg/L	EPA 8081	03/31/17	GMB	TC regulatory limit is 0.02 mg/L.
Heptachlor	<0.004	mg/L	EPA 8081	03/31/17	GMB	TC regulatory limit is 0.008 mg/L.
Heptachlor epoxide	<0.004	mg/L	EPA 8081	03/31/17	GMB	TC regulatory limit is 0.008 mg/L.
Lindane by ECD	<0.004	mg/L	EPA 8081	03/31/17	GMB	TC regulatory limit is 0.4 mg/L.
Methoxychlor	<0.02	mg/L	EPA 8081	03/31/17	GMB	TC regulatory limit is 10.0 mg/L.
Toxaphene	<0.04	mg/L	EPA 8081	03/31/17	GMB	TC regulatory limit is 0.5 mg/L.
TC Herbicides	See below		EPA 8151	04/03/17	GMB	
2,4,5-TP (Silvex)	<0.2	mg/L	EPA 8151	04/03/17	GMB	TC regulatory limit is 1.0 mg/L.
2,4-D	<0.2	mg/L	EPA 8151	04/03/17	GMB	TC regulatory limit is 10.0 mg/L.

Attest: 
David R. Alkema, Lab Manager

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **703913**

Date Reported : **04/12/17**

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : <u>ZHE Leachate of Digester #4</u>						
Sampled By :			Date Received : 03/29/17			
Sample Date :			Sample Type : ZHE			
Sample Time :			KAR Sample No. : 703913-01Z			
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Volatiles	See below		EPA 8260	04/03/17	JAR	
Prep. VOA	Completed		EPA 5030	04/03/17	JAR	
1,1-Dichloroethene	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 0.7 mg/L.
1,2-Dichloroethane	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 0.5 mg/L.
1,4-Dichlorobenzene	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 7.5 mg/L.
Benzene	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 0.5 mg/L.
Carbon tetrachloride	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 0.5 mg/L.
Chlorobenzene	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 100 mg/L.
Chloroform	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 6.0 mg/L.
Methyl ethyl ketone	<2	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 200 mg/L.
Tetrachloroethene	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 0.7 mg/L.
Trichloroethene	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 0.5 mg/L.
Vinyl chloride	<0.1	mg/L	EPA 8260	04/03/17	JAR	TC regulatory limit is 0.2 mg/L.

Attest: 
David R. Alkema, Lab Manager

KAR Laboratories, Inc.

February 21, 2017